Conclusion: Evidence Landscape, Methodological Limitations, and Research Agenda

Summary I

This work demonstrates a first-generation prototype infrastructure for citation-weighted evidence mapping and hypothesis triage across a rapidly growing scientific field. By combining multi-source retrieval ($N = 817$ papers from three databases, inclusion from 2000 onward), LLM-based assertion extraction encoded as provenance-bearing nanopublications, and citation-weighted triage scoring, we produce a queryable, RDF-compatible knowledge graph that maps the evolving evidence landscape for eight core Active Inference claims. The system demonstrates the feasibility of automated living reviews, while clearly delineating the boundaries of current model capabilities.

Summary II

All assertions and hypothesis scores are machine-generated; outputs always report evidence mapping and triage, not scientific confirmation—there is no live gate that suppresses or changes output based on the validation metrics below, which are reported as reproducibility context, not an enforced pass/fail threshold. A stratified rule-based reference-annotator agreement study ($n = 200$) yields inter-rule $\kappa = 0.554$ and pipeline precision 0.000, recall 0.000 against a deterministic keyword-rule reference; direction agreement between that reference and the pipeline is only $\kappa = -0.055$. These are reproducibility signals, not accuracy against human labels—the pre-declared human gold-standard baseline below has not yet been collected.

Constraints and Methodological Scope I

Several conscious design constraints scope these findings.

Keyword Classifier Resolution

The keyword-based classifier operates over 200+ keyword indicators distributed across 8 domain categories (74 mathematical indicators in A1 alone), using a deterministic priority system that routes papers to specific application domains (C1–C5) before testing tools (B), formal theory (A1), and the qualitative philosophy catch-all (A2). Word-boundary-aware matching reduces partial-match false positives, but keyword-based methods cannot capture semantic nuance: papers using novel terminology or discussing cross-domain topics without standard vocabulary risk misclassification. Residual A2 concentration should be interpreted as a ceiling on broad theoretical generality rather than a literal measure of philosophical focus. An embedding-based classifier trained on a labeled subset would provide a quantitative upper bound on the fraction of A2 papers that merit redistribution.

Constraints and Methodological Scope II

Citation Network Coverage Gaps

The 2,176 intra-corpus edges spanning 547 connected components provide a topological skeleton, but three systematic gaps inflate the component count: (1) cross-source identifier mismatches (DOI vs. OpenAlex vs. arXiv ID), (2) papers whose references are not indexed by any source API, and (3) open-access preprints whose DOIs differ from their published versions. Exhaustive DOI-level cross-matching with fuzzy title matching would condense the graph further.

Constraints and Methodological Scope III

Corpus Biases, Citation Dynamics, and Linguistic Framing

Citation counts are subject to Matthew effects and cumulative field-size biases. Partial-year indexing for the most recent calendar year undercounts recent publications. The measured 20.36% CAGR reflects the dilutive effect of the long temporal span (2005–2026), corresponding to a 2.8-year publication doubling time; the growth phase from 2010 onward follows a steeper trajectory with a substantially shorter doubling interval. Additionally, the retrieved corpus itself suffers from selection biases inherent to queried databases, including English-language dominance and the structural over-indexing of preprints relative to peer-reviewed final versions. Finally, the predominantly positive hypothesis scores across the board are inflated by two systematic effects: (1) **publication bias**, which causes academic journals to preferentially select positive and confirmatory findings [sterling1959publication], and (2) **linguistic asymmetry** in scientific writing, where declarative claims are phrased affirmatively far more often than negatively. These effects jointly suppress contradicting assertions in the extracted evidence base. Relative rankings and temporal

Research Agenda: Four Priority Next Steps I

The current prototype establishes a reproducible baseline and surfaces the field's evidence structure at corpus scale. Four concrete next steps, ordered by the dependency chain each one unlocks, define the path from prototype to production-grade living review.

Research Agenda: Four Priority Next Steps II

Next Step 1 — Expand the Scope of Referenced Data

The present corpus of $N = 817$ papers is retrieved via keyword queries against three APIs (Semantic Scholar, OpenAlex, arXiv). Three expansion axes would materially change the evidence landscape.

Additional sources. PubMed, PsycINFO, and IEEE Xplore each index Active Inference literature that the current APIs do not reach: neuroscience clinical trials (PubMed), cognitive-behavioral studies (PsycINFO), and robotics control architectures (IEEE). For each new source, the retrieval layer requires only a source-specific connector implementing the same `fetch_papers(query, max_results)` interface used by existing adapters. Gray literature—technical reports, theses, and institutional preprints not yet indexed by major APIs—represents an additional tier: harvesting from ORCID work records and institutional repositories would capture practitioner findings that never appear in indexed venues.

Future Directions: Beyond Tally-Based Evidence Aggregation I

Beyond the four priority next steps above, the scoring machinery itself can be upgraded. We identify four directions, ordered by expected impact.

Future Directions: Beyond Tally-Based Evidence Aggregation II

Hierarchical Bayesian Hypothesis Scoring

The most direct extension replaces the additive tally with a **hierarchical Bayesian model** that treats each hypothesis score as a latent variable inferred from noisy assertion observations. Under this formulation, each assertion a_i contributes a likelihood term $P(a_i|\theta_H, \sigma)$ parameterized by the hypothesis-level evidence strength θ_H and an observation noise term σ capturing LLM extraction uncertainty. A hierarchical prior $\theta_H \sim \mathcal{N}(\mu_{\text{field}}, \tau^2)$ pools information across hypotheses, enabling principled shrinkage for hypotheses with sparse evidence (e.g., H6 Clinical Utility, with only 38 assertions). This framework produces posterior credible intervals rather than point estimates, providing uncertainty quantification that the current tally-based scores lack. Temporal dynamics can be modeled through time-varying parameters $\theta_H(t)$ using state-space formulations that re-weight older evidence rather than treating all cumulative assertions equally.

Future Directions: Beyond Tally-Based Evidence Aggregation III

Causal Evidence Graphs

A second-generation knowledge graph would encode not only assertion-level relationships (paper $\rightarrow$ supports $\rightarrow$ hypothesis) but also **causal dependencies among hypotheses** themselves. For example, evidence for predictive coding (H4) often implicitly supports FEP universality (H1), yet the tally-based approach treats them as independent. A causal evidence graph—structured as a directed acyclic graph (DAG) over hypotheses with edge weights learned from co-assertion patterns—would enable cross-hypothesis evidence propagation using belief propagation or variational message passing. This is particularly relevant for the Active Inference literature, where hypotheses are theoretically nested: FEP universality (H1) logically entails predictive coding (H4), and Markov blanket realism (H3) is a prerequisite for certain formulations of H1. Encoding these dependencies would prevent the double-counting of evidence from papers that support multiple related hypotheses and enable identification of which specific

Limitations I

Three constraints bound the current findings. First, extraction operates on abstracts only: full-text methods, results, and supplementary data—where quantitative effect sizes and experimental controls live—are not yet parsed. The rubric's evidence-quote fidelity dimension (Step 4, Table~**??**) will quantify exactly how much signal this omission suppresses once a full-text pilot is run. Second, keyword-based retrieval across three APIs produces a snapshot with systematic false negatives: papers using non-canonical terminology, gray literature, and domain-adjacent work (EBM, Bayesian brain models) are undercounted. The corpus recall metric provides a principled bound on this gap rather than a vague acknowledgement of it. Third, the citation-weighted tally treats all assertion sources symmetrically; the evidential diversity and outcome-grounding extensions above are the concrete remedies. These are not general disclaimers but tracked deficits against which the Step 1–4 roadmap makes measurable progress.

Broader Impact I

Knight et al. [knight2022fep] identified three capabilities as goals for the field: “encompass increased scope of relevant works,” “integrate multiple forms of annotation and participation,” and “facilitate integration of manual and artificial contributions.” The four-step research agenda in §**??** operationalizes each of these directly: Step 1 addresses scope, Step 2 addresses the quality of extracted contributions, Step 3 addresses empirical grounding, and Step 4 provides the formal rubric that makes “integration of manual and artificial contributions” verifiable rather than aspirational.

Broader Impact II

By demonstrating that LLM-driven assertion extraction can produce scalable, queryable representations of scientific evidence—processing $N = 817$ papers spanning approximately two and a half decades (2005–2026), extracting structured assertions, and evaluating 8 core hypotheses—this work provides a reusable architecture for realizing this vision. The corpus window begins in 2005 to capture Energy-Based Model and variational Bayesian antecedents that predate the Free Energy Principle label itself; the formal FEP was introduced in 2006 [friston2006free] and reached its core elaboration by 2010 [friston2010free]. The citation network metrics (2,176 edges, 0.33% density, mean in-degree 2.7) characterize the field's structure, which has grown at a 20.36% CAGR while diversifying across 5 application domains.

Broader Impact III

The limitations of keyword-based retrieval across disjoint academic repositories mean that any retrieved corpus will contain both false positives and false negatives. There is no single threshold that perfectly defines inclusion or exclusion for a dynamic, interdisciplinary research field. The primary contribution of this work is therefore not a definitive corpus but an open-source, modularly updatable, and versioned software package. This tool is built in reference to custom literature bibliographies that can be iteratively curated for relevance by the community.

Broader Impact IV

The combination of multi-source retrieval, LLM-based extraction, and probabilistic knowledge graph construction provides a reusable template that advances each of these goals. A complementary pathway is emerging through Retrieval-Augmented Generation (RAG) architectures that ground LLMs directly in knowledge graphs, reducing hallucination and enabling real-time, context-aware reasoning over structured evidence [fan2024survey]. Integrating our nanopublication graph into such a RAG system would enable natural-language querying of the evidence base, further lowering the barrier for community engagement. The recent release of nanopub-js v0.1.0 [kuhn2026nanopubjs]—enabling browser-based creation, signing, and querying of nanopublications—lowers the barrier for community-contributed assertions, bringing the participatory evidence curation envisioned by Knight et al. within practical reach. As LLM capabilities improve and standardized metadata adoption grows, the cost of maintaining such systems will decrease while their utility increases. By open-sourcing the pipeline and publishing the schema, we provide both a concrete tool for the Active Inference community